Riesz representation Theorem.

Thm. Let V be a fin. dim. inn. prod. v.s.
For any linear functional $f: V \to F$, there is a unique vector $\beta \in V$ s.t.

$$\text{for all } \alpha \in V, \quad f(\alpha) = \langle \alpha, \beta \rangle$$

In other words, $\Phi: V \to V^*: \beta \mapsto \langle \cdot, \beta \rangle$ is a Conjugate isomorphism.

Pf. Let $\{\alpha_1, \cdots, \alpha_n\}$ be an orthonormal basis for V.
Given linear functional $f: V \to F$, put

$$\beta = \sum_{j=1}^{n} \overline{f(\alpha_j)} \cdot \alpha_j.$$

then, for each $1 \le k \le n$,

$$\langle \alpha_k, \beta \rangle = \left\langle \alpha_k, \sum_{j=1}^{n} \overline{f(\alpha_j)}\, \alpha_j \right\rangle = \sum_{j=1}^{n} f(\alpha_j) \cdot \langle \alpha_k, \alpha_j \rangle = f(\alpha_k)$$

Clearly, $\langle \cdot, \beta \rangle$ is linear, thus the existence is proved. And the uniqueness is shown:
if $\langle \alpha, \beta \rangle = \langle \alpha, \gamma \rangle$ for all $\alpha \in V$, then for $\alpha = \beta - \gamma$, $\langle \beta - \gamma, \beta - \gamma \rangle = 0$.

Thus, $\beta = \gamma$. ☒

Observe that a inn. prod space V with orthonormal basis $\{\alpha_1, \cdots, \alpha_n\}$
Let $\alpha = x_1\alpha_1 + \cdots + x_n\alpha_n$ and $\beta = y_1\alpha_1 + \cdots + y_n\alpha_n$. Then,

$$\langle \alpha, \beta \rangle = x_1\overline{y_1} + \cdots + x_n\overline{y_n}$$

Meanwhile, for any linear functional $f \in V^*$,

$$f(\alpha) = f(\alpha_1)x_1 + \cdots + f(\alpha_n)x_n$$

If we want to β satisfies $\langle \alpha, \beta \rangle = f(\alpha)$, then take $y_i = \overline{f(\alpha_i)}$. That is,

$$\beta = \overline{f(\alpha_1)}\,\alpha_1 + \cdots + \overline{f(\alpha_n)} \cdot \alpha_n$$

Riesz-Representation theorem

Thm. Let V be a finite-dimensional inner product space over F. Given a linear function $f: V \to F$, there exists a unique vector $\beta \in V$ such that $f(\alpha) = \langle \alpha, \beta \rangle$ for all $\alpha \in V$.

In other words, $\Phi: V \to V^*: \beta \mapsto (\alpha \mapsto \langle \alpha, \beta \rangle)$ is a conjugate isomorphism.

Proof. By the Gram-Schmidt process, let $\{\alpha_1, \cdots, \alpha_n\}$ be an orthonormal basis for V. Let $f: V \to F$ be a linear functional. Put

$$\beta = \sum_{k=1}^{n} \overline{f(\alpha_k)} \cdot \alpha_k$$

Then, given $1 \le i \le n$,

$$\langle \alpha_i, \beta \rangle = \langle \alpha_i, \sum_{k=1}^{n} \overline{f(\alpha_k)} \cdot \alpha_k \rangle = \sum_{k=1}^{n} f(\alpha_k) \cdot \langle \alpha_i, \alpha_k \rangle = f(\alpha_i)$$

So that $f(\alpha_i) = \langle \alpha_i, \beta \rangle$.

[Geometric Observation.
Given a linear functional $f: V \to F$, $V = \ker f \oplus (\ker f)^{\perp}$.
Let P be the orthogonal Projection of V on $(\ker f)^{\perp}$. Then,

for all $\alpha \in V$, $f(\alpha) = f(P(\alpha))$.

Now, if $f \neq 0$, then $\operatorname{rank} f = 1$, so for any non-zero vector $\gamma \in (\ker f)^{\perp}$,

$$P(\alpha) = \frac{\langle \alpha, \gamma \rangle}{\|\gamma\|^2} \cdot \gamma \quad \text{for all } \alpha \in V.$$

Finall, for given $\alpha \in V$,

$$f(\alpha) = f(P(\alpha)) = \langle \alpha, \gamma \rangle \cdot \frac{f(\gamma)}{\|\gamma\|^2}, \quad \text{and } \beta = \frac{\overline{f(\gamma)}}{\|\gamma\|^2} \cdot \gamma.$$